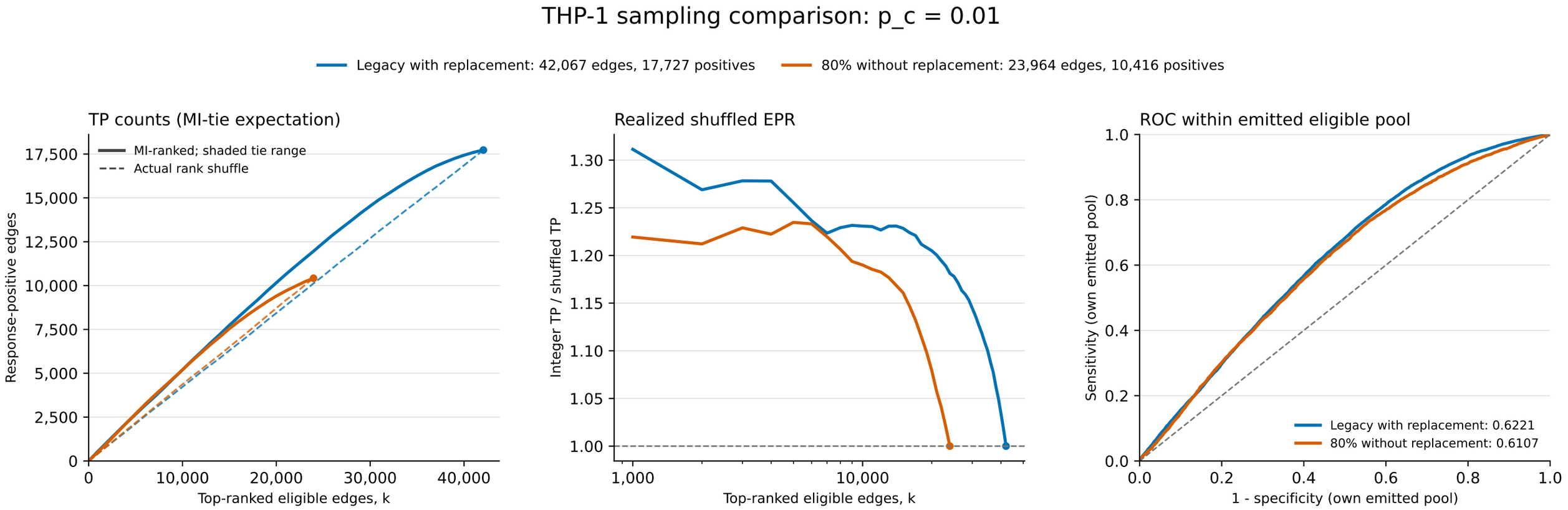
THP-1 sampling comparison: p_c = 0.01
Legacy with replacement: 42,067 edges, 17,727 positives
80% without replacement: 23,964 edges, 10,416 positives
TP counts (MI-tie expectation)
MI-ranked; shaded tie range
Actual rank shuffle
Response-positive edges
17,500
15,000
12,500
10,000
7,500
5,000
2,500
0
0
10,000
20,000
30,000
40,000
Top-ranked eligible edges, k
Realized shuffled EPR
Integer TP / shuffled TP
1.30
1.25
1.20
1.15
1.10
1.05
1.00
1,000
10,000
Top-ranked eligible edges, k
ROC within emitted eligible pool
Sensitivity (own emitted pool)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
1 - specificity (own emitted pool)
Legacy with replacement: 0.6221
80% without replacement: 0.6107
Each arm stops at its own emitted pool. EPR uses its own realized R seed-1 shuffle; ROC retains MI ties.